

Birla Institute of Technology & Science, Pilani, Rajasthan

First Semester 2021-2022 Mid-Semester Exam (Open Book)

EEE F311 Communication Systems

Date: 30-08-2021, Duration: 90 Minutes, Maximum Marks: 50

- There are 5 questions each with 10 marks.
 - Only textbook and handwritten notes are allowed.
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- (a) Which of the following codes are uniquely decodable with justification? $C_1 = \{00, 01, 0\}$, $C_2 = \{00, 01, 100, 101, 11\}$, $C_3 = \{0, 10, 110, 1110, \dots\}$, $C_4 = \{0, 00, 000, 0000\}$.
Solution: C_2, C_3 : Satisfies the prefix condition.

(b) An amplitude modulated signal is given by $(A_c + A_m m_n(t)) \cos 2\pi 10^3 t$, where $m_n(t)$ denotes the normalized message signal. Find the minimum carrier power required to make envelope detector working with probability one if $A_m \sim U(-6, 6)$, where U denotes the uniform random variable.
Solution: Carrier Power = $\frac{6^2}{2} = 18$.

(c) A signal $m(t) = 2\text{sinc}(2\pi t)$ is passed through a system with impulse response $g(t) = \frac{1}{\pi(t-5)}$. Find the energy of the signal at the output.
Solution: Energy of input signal = 2. Hilbert filter does not change the energy. Thus, the output energy is 2.

(d) A wireless channel has an specification: bandwidth 100 Hz at a carrier frequency of 1000 Hz. Draw a transmitter for DSB-SC modulation if the transmitted signal is $m(t) = \Pi(t/10)$, where $\Pi(\cdot)$ denote the rectangular pulse.
Solution: The message signal is passed through a low pass filter $g(t) = 100\text{sinc}(100\pi t)$ (50 Hz channel because BW gets doubled for DSB-SC) multiplied with $A\cos(2\pi 1000t)$.

(e) In a broadcast radio system, there are 3 stations with messages $m_1(t)$, $m_2(t)$, and $m_3(t)$ operating in a region with carrier frequencies 100 Hz, 120 Hz, and 130 Hz, respectively. The bandwidth of each message is 5 Hz. A superhetrodyne receiver is used to detect using high-end tuning of the local oscillator and IF amplifier of 10 Hz bandwidth. The front-end BPF has an specification of 200 Hz bandwidth at a center frequency 100 Hz. How many signals will be received at the output of IF amplifier if the radio is tuned at 100 Hz.
Solution: 2: Both $m_1(t)$ and image signal $m_2(t)$ for $f_{IF} = 10$ Hz.
2. Consider a periodic message signal as $m(t) = \triangle(t/2)$ and the carrier signal $10\cos(2\pi 10^4 t)$.

- (a) Find the transmitted power for the DSB-SC.

Solution: We draw a triangular pulse in positive amplitude direction. The equation for triangular pulse is $m(t) = t + 1$ for $t = -1$ to 0 , and power $P_m = \int_{-1}^0 (t + 1)^2 dt = \frac{1}{3}$, thus transmitted power for DSB-SC is equal to $A^2 P_m / 2 = 100/6$.

- (b) Find the modulation index of the conventional AM signal (i.e., DSB-SC with carrier).

Solution: Modulation index is $1/10$.

- (c) Plot the U-SSB signal.

Solution: A sinc square pulse with positive amplitude from 10 KHz to ∞ . Symmetrical plot at -10 KHz.

- (d) Find the maximum frequency deviation for the FM signal with $k_f = 10$ Hz/Volt.

Solution: $m_p = 1$, Thus $\Delta f = k_f m_p = 10 \times 1 = 10$.

- (e) Find the maximum frequency deviation for the PM signal with $k_p = \pi$ rad/Volt.

Solution: $\Delta f = \frac{1}{2\pi} k_p \frac{dm(t)}{dt} = 1/2$ since maximum of $\frac{dm(t)}{dt} = 1$.

3. (a) A digital storage device needs to store two message with probabilities $P_1 = 1/3$ and $P_2 = 2/3$ as binary digits. Find the binary code of each message and code efficiency. How many order extension is required to achieve a coding efficiency greater than 95% with Shannon-Fano variable-length encoding. Find the codes of each extended message and coding efficiency achieved.

Solution: Binary code 0, 1, length =1, entropy: 0.9183. Efficiency 0.9183. 2-order extended code 0, 10, 110, 111. Average length= 0.9444. Efficiency= 0.9723

- (b) Find the 4-ary Huffman code and coding efficiency for messages with probabilities $m_1 = 6/25$, $m_2 = 6/25$, $m_3 = 4/25$, $m_4 = 4/25$, $m_5 = 3/25$, $m_6 = 2/25$.

Solution: The codes are 0,1,2, 30, 31, 32. Average length = 1.36. Entropy = $-(p_1 \log_2(p_1) + p_2 \log_2(p_2) + p_3 \log_2(p_3) + p_4 \log_2(p_4) + p_5 \log_2(p_5) + p_6 \log_2(p_6)) = 2.4929/(\log_2 4) = 1.2464$.

4. (a) A receiver has many amplifier stages. Consider three amplifier stages A, B, and C with specifications: A(gain 20 dB, noise figure 8 dB), B(gain 30dB, noise figure 5dB), C(gain 10dB, noise figure 10dB). Arrange these amplifier in a cascaded order to maximize the overall noise figure and find the equivalent noise temperature of the cascaded system.

Solution: CAB: Over noise figure = $F = F_C + (F_A - 1)/(G_C) + (F_B - 1)/(G_C G_A) = 10.5331$
Noise temperature: $T_e = (F - 1)T_0 = (10.5331 - 1) = 2764\text{K}$.

- (b) A zero-mean white Gaussian noise $n(t)$ with double-sided power spectral density 10 W/Hz is passed through a low pass filter of bandwidth 100 Hz, as shown in the figure. Find the auto-correlation function of the output signal $x(t)$.

Solution: $S_X(f) = S_N(f)H(f)^2$ with $H(f) = \Pi(\frac{f}{200})$. $S_X(f) = 10\Pi(\frac{f}{200})$. Taking inverse Fourier transform, we obtain the autocorrelation function of the output $R_X(\tau) = 2000\text{sinc}(200\pi\tau)$.



5. An FM signal $x(t) = 20 \cos(2\pi 10^5 t + 0.3 \sin 2\pi 100 t)$ is passed through multipliers M_1 and M_2 , Local oscillators with frequencies f_{LO1} and f_{LO2} , and bandpass filter BPF 1 and BPF 2 to generate another FM signal $y(t) = 20 \cos(2\pi 10^7 t + 15 \sin 2\pi 10 t)$. Find the specifications of M_1 , M_2 , f_{LO1} , f_{LO2} ,

and bandwidth and carrier frequencies for BPF 1 and BPF 2. Design should have constraints on multipliers $M_1 > 1$, $M_2 > 1$, and local oscillators f_{LO1} , f_{LO2} having a maximum frequency of 10 MHz.

Solution: Multiple solutions depending on choice of M_1 and M_2 . We can take $M_1 = 25$ and $M_2 = 2$, which gives $\beta = 15$. The carrier frequency after $M_1 = 100\text{KHz} \times 25 = 2.5\text{MHz}$ and $\beta = 0.3 \times 25 = 7.5$. Bandwidth of the message signal using carson's rule = 1.7KHz. We take L_{O1} at 1.5MHz. BPF 1: BW= 1.7 KHz, center frequency 6 MHz. Carrier after M_2 : 12 MHz. We take L_{O2} at 2MHz. BPF 1: BW= 3.2 KHz, center frequency 10 MHz.

